

PROJECT ANNOUNCEMENT
Fall 2026

**Design, Construction and Testing of an Unmanned
Vertical Takeoff and Landing Aircraft**

Position Ref : VTOL
Type : Group
Status : New
Seats : 2 Students (Muhammed Yusuf Özeroğlu,
Mahir Deniz Çevik)



Project Description:

The project involves design, construction and testing of a model airplane that can takeoff and land vertically. Three alternative configurations will be studied.

i) The engines can be rotated by 90° relative to the wing to facilitate liftoff and landing while the airplane body is in horizontal position. The engines would provide vertical thrust for liftoff, which would then be rotated to produce horizontal thrust for level flight. Similar steps would be followed for vertical landing.

ii) The whole wing can be rotated by an angle somewhat less than 90° with the engines remaining stationary relative to the wing while the airplane body is in horizontal position.

iii) The whole airplane can sit on its tail in a vertical position for liftoff. Once the cruise/mission altitude is reached the whole aircraft would turn by 90° for level flight. The wing and the engines would be stationary relative to the airplane body.

Work Description:

- Literature survey on VTOL aircraft.
- Design of a two-engine VTOL aircraft.
- Determination of the mass center.
- Estimation of the aerodynamic center.
- Estimation of flight loads and derivation of flight mechanics equations.
- Estimation of the static margin.
- Investigation of stability during vertical-to-level flight transition.
- Construction of the airplane.
- Testing the airplane.

Required Qualifications and Skills:

- Appetite for theoretical and applied work.
- Appetite to learn basic aerodynamics and static flight stability.
- The project will be on campus.

Related UN Sustainable Development Goals:

- No.9 Industry, innovation and infrastructure.

Difficulty: high (X) medium () low ()

Time commitment: high (X) medium () low ()

Estimated expense: 12.000,00 TL